WEEK 2 · DAY 3 · OVERVIEW

Clean and Dirty Price



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ABOUT 45 MIN WITH EXERCISES / 9 SECTIONS / 4 EXERCISES

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The Desk Lead's Note

We buy 1,000,000 of Quorvane Energy at its issue yield, 5.40 percent, settling Monday 30 November. Tell me the quote and the cash that leaves the account. Then give bondmath a price for any settlement date and a yield from any quote, both tied to Excel. Last, September's blotter shows principal only: put the accrued back on all 374 trades and tell me what the cash really was.

ABOUT 45 MIN WITH EXERCISES / 9 SECTIONS / 4 EXERCISES

Your Route Through Day 3

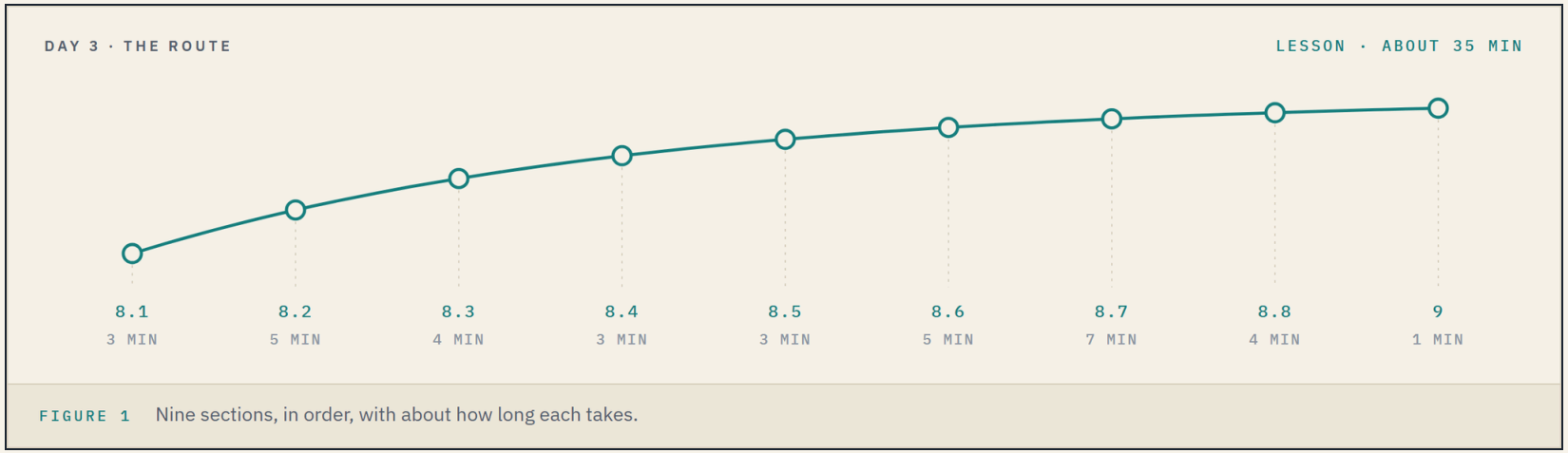


FIGURE 1 Nine sections, in order, with about how long each takes.

	SECTION	ABOUT
8.1	The Quote and the Invoice	3 min
8.2	Discounting From Settlement, `w = DSC / E`	5 min
8.3	**`price`** and **`dirty_price`**	4 min
8.4	On a Coupon Date It Equals Notebook 02	3 min
8.5	The Final Coupon Period	3 min

	SECTION	ABOUT
8.6	Undo a Commit, Keep the History	5 min
8.7	**`yield_to_maturity`**	7 min
8.8	The September Blotter's Invoices	4 min
9	Bring It Home	1 min

Time. About 35 minutes for the lesson, then about 12 for the four exercises.

8.1 The Quote and the Invoice

THE FOUR BONDS

	ISSUER	COUPON_PCT	MATURITY	YIELD_PCT
0	Quorvane Energy	5.250	2031-09-30	5.400
1	Dunmarrow Rail	6.125	2033-09-30	6.125
2	Pellwick Paper	4.750	2029-09-30	4.600
3	Halvering Foods	7.500	2036-09-30	7.700

QUOTE AND ACCRUED

```
issue_price = bondmath.price_from_yield(5.25, 5.40, 5)
print(f'Price on 30 September: {issue_price:.6f}')
print(f'Accrued on 30 November: {quorvane_accrued}')
```

```
Price on 30 September: 99.350327
Accrued on 30 November: 0.875
```

THE TERMS

The market quotes the **clean** price; the buyer pays the **dirty** price, clean plus accrued (notebook 02's terms).

The **invoice** is the cash that settles the trade: the dirty price applied to the face.

8.2 Discounting From Settlement, w = DSC / E

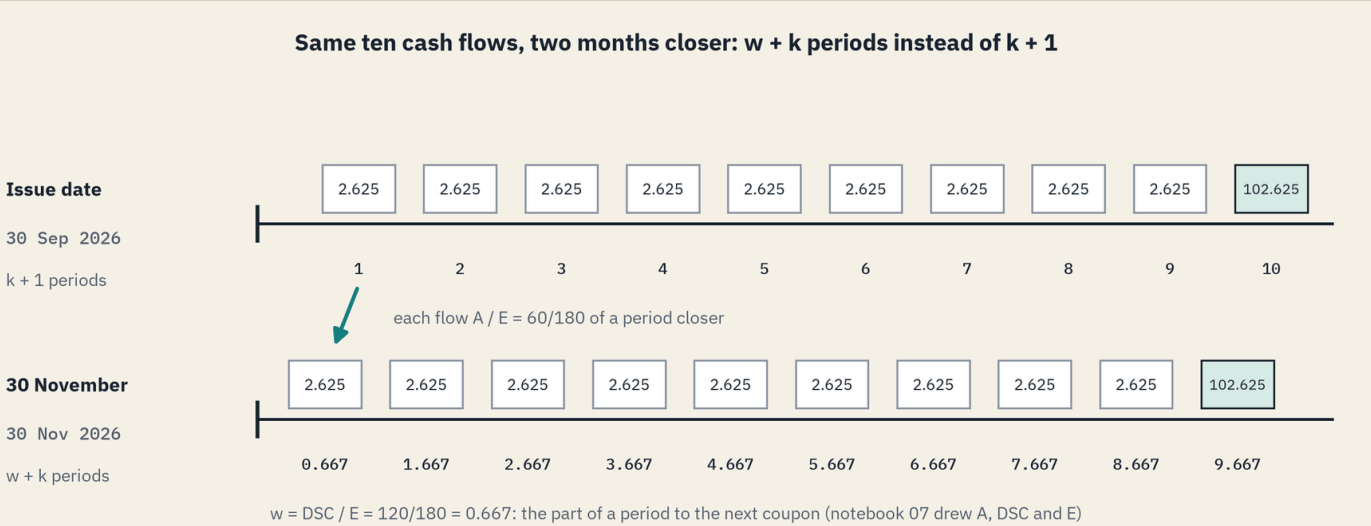


FIGURE 2 Quorvane's ten cash flows sit 0.667 to 9.667 periods from 30 November, a third of a period nearer than at issue.

A, E, DSC, AND W

```
w = days_to_next / days_in_period

A = 60, E = 180, DSC = 120, w = 0.666667
```

THE CASH FLOWS

	PERIODS	CASH_FLOW	PRESENT_VALUE
0	0.666667	2.625	2.578788
1	1.666667	2.625	2.510991
2	2.666667	2.625	2.444977
3	3.666667	2.625	2.380698

The dirty price is the cash flows discounted from settlement; the quote is that less accrued.

8.3 price and dirty_price

THE LOOP INSIDE PRICE

```
    w = days_to_next / days_in_period
    period_yield = yield_pct / 100 / frequency
    if coupons_left == 1:
        dirty = (100 + coupon) / (1 + w * period_yield)
        dirty = 0.0
        for k in range(coupons_left):
            dirty += cash_flow / (1 + period_yield) ** (w + k)
```

TESTED AGAINST EXCEL

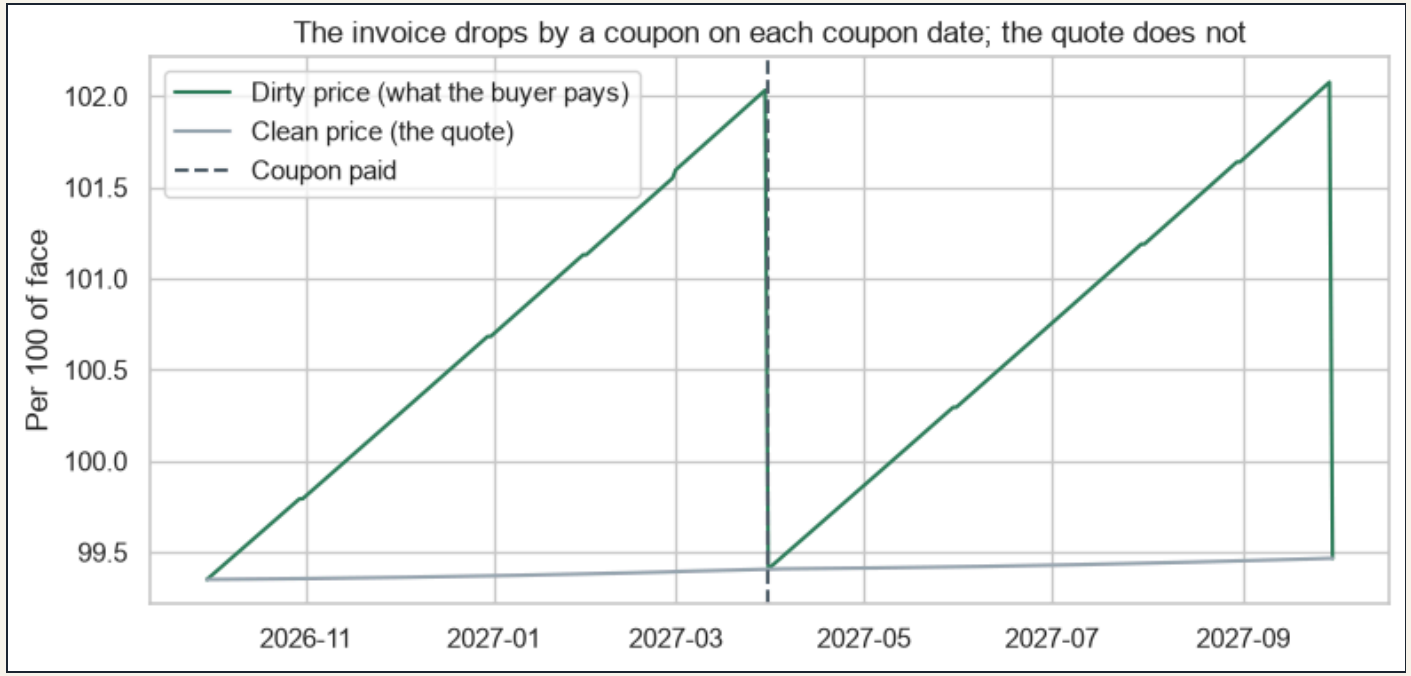
```
def test_price_matches_excel():
    for row in REFERENCE:
        result = bondmath.price(*bond_inputs(row))
def test_dirty_is_clean_plus_accrued():
    for row in REFERENCE:
        result = bondmath.dirty_price(*bond_inputs(row))
```

DIRTY_PRICE ADDS THE ACCRUED BACK

```
    return (price(settlement, maturity, coupon_pct, yield_pct, frequency, basis)
            + accrued_interest(settlement, maturity, coupon_pct, frequency, basis))
```

`price` **matches Excel's** `PRICE`**, and** `dirty_price` `PRICE` **plus accrued, on all 306 reference rows.**

8.4 On a Coupon Date It Equals Notebook 02



THE COUPON DATE

```
Dirty on 30 and 31 March 2027: 102.032786, 99.407786
```

WHAT CHANGES ON 31 MARCH

The dirty price falls by exactly 2.625 on 31 March: under 30/360, 30 March already counts all 180 days.

The new function keeps every old answer, and the quote does not jump on coupon day.

8.5 The Final Coupon Period

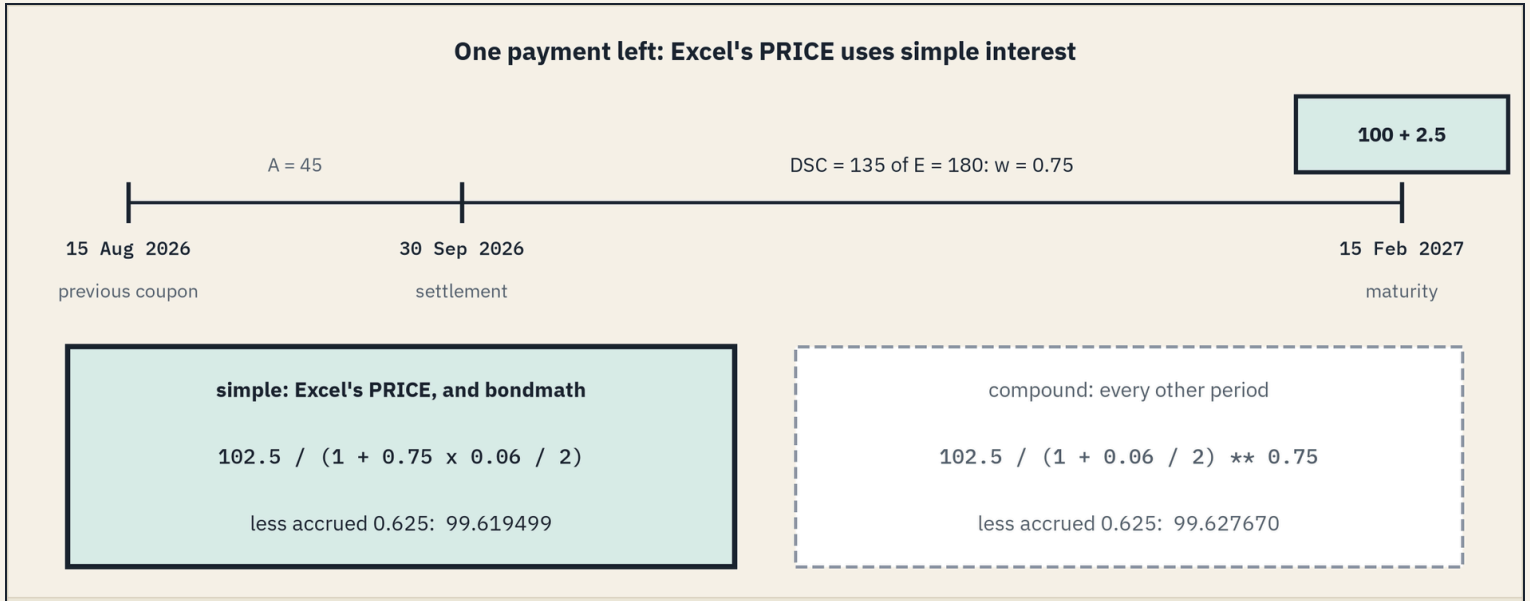


FIGURE 3 With one coupon left, simple interest gives 99.619499; compounding would give 99.627670.

SIMPLE AGAINST COMPOUND

```
w = 0.75; Excel's PRICE 99.6194987775
Simple:   99.6194987775
Compound: 99.6276697643
```

THE COST

price is Excel's number to the last digit; compounding would add 0.008171 per 100, 81.71 dollars on 1,000,000.

With one coupon left, Excel uses simple interest, and so does `bondmath`.

8.6 Undo a Commit, Keep the History

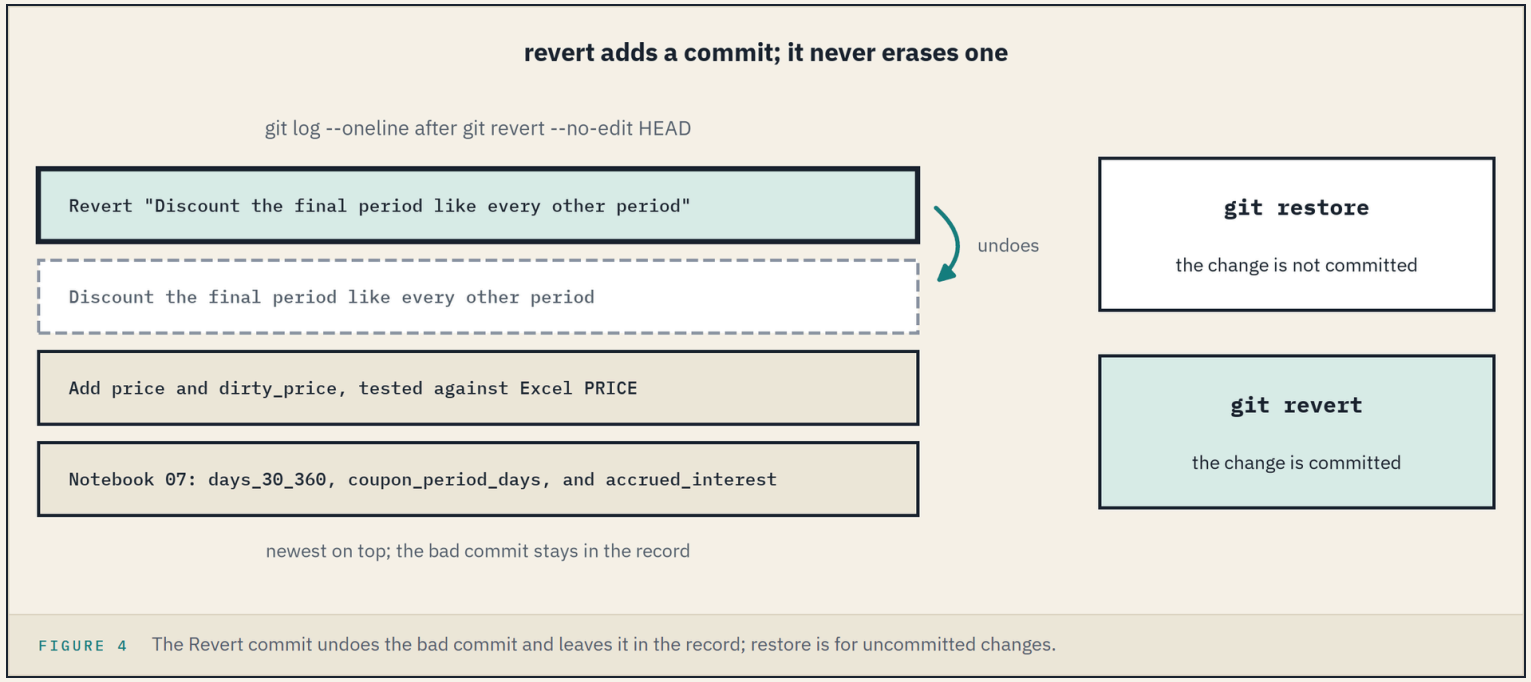


FIGURE 4 The Revert commit undoes the bad commit and leaves it in the record; restore is for uncommitted changes.

UNDO IT WITH A NEW COMMIT

```
!git -C "{work}" revert --no-edit HEAD
!git -C "{work}" log --oneline -4
```

REVERT, NOT RESTORE

`git revert` makes a new commit that does the opposite of an old one.

`restore` undoes what is not committed; `revert` undoes what is and keeps the record. Never rewrite a shared history; add to it.

8.7 yield_to_maturity

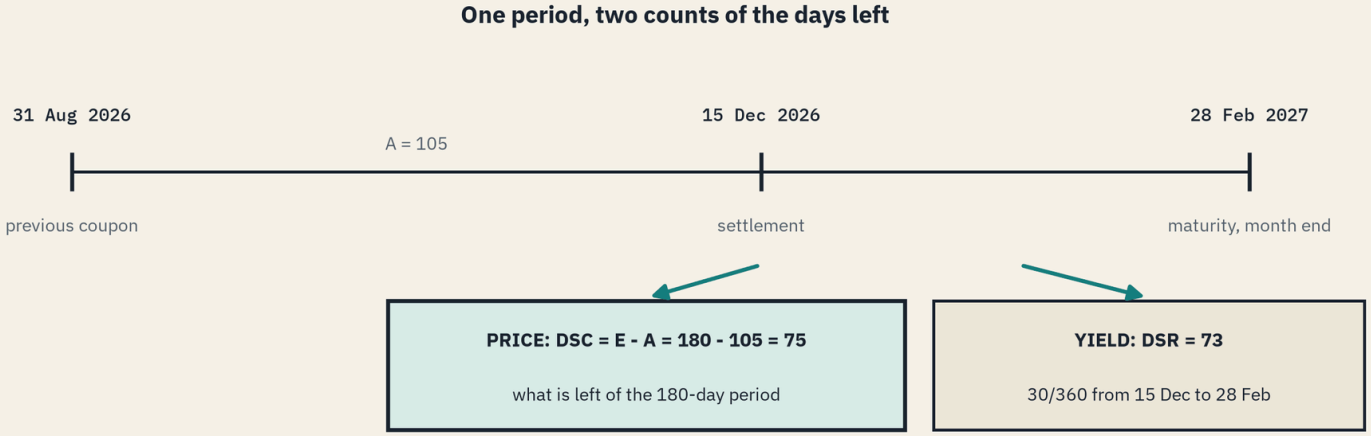


FIGURE 5 PRICE counts 75 days left to a 28 Feb maturity; YIELD counts 73 on 30/360.

THE ISSUE YIELDS COME BACK

```
Largest gap from the issue yields: 1.9e-11 percent
```

SEVENTY-FIVE DAYS OR SEVENTY-THREE

```
E - A for PRICE: 75; DSR for YIELD: 73
```

From any quote to its yield, as Excel's `YIELD`, final period included.

8.8 The September Blotter's Invoices

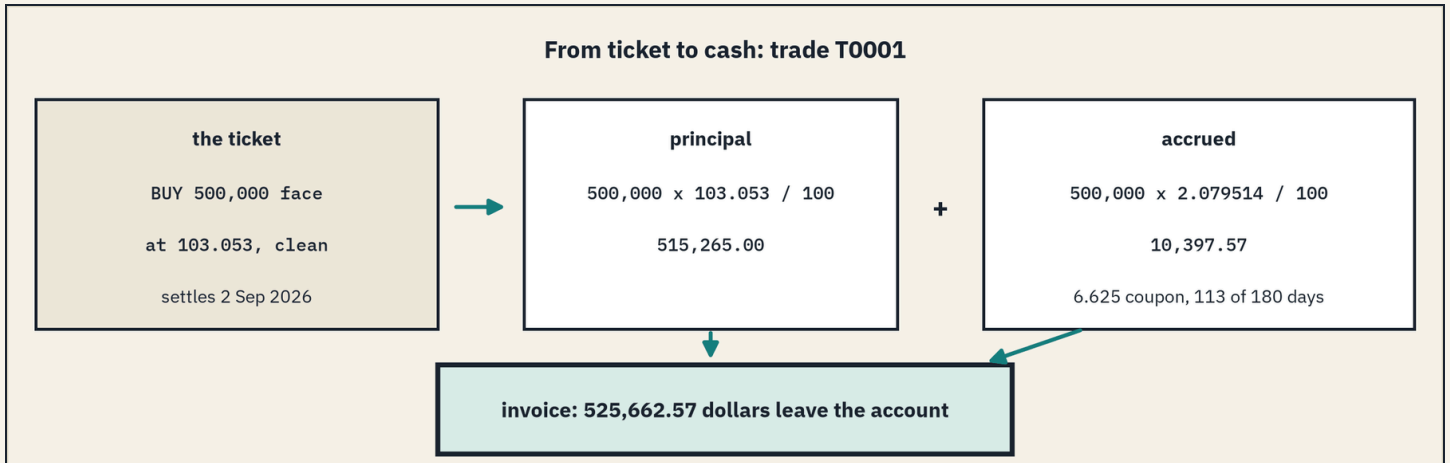


FIGURE 6 Trade T0001 settles for 525,662.57 dollars: 515,265.00 of principal plus 10,397.57 of accrued.

EVERY TRADE

```
2 trades accrued nothing; the most accrued: T0151
```

BY SIDE

SIDE	ACCRUED_DOLLARS	INVOICE_DOLLARS
BUY	990,620.95	57,546,626.40
SELL	819,168.26	61,736,912.16
All	1,809,789.22	119,283,538.57

September's trades settled for 119,283,538.57 dollars: 1,809,789.22 of accrued sits outside the principal column.

Bring It Home

IN A POWERSHELL TERMINAL

```
Set-Location "$HOME\projects\my_bondmath"
python -m pytest -q
git diff --stat
git commit -am "Add price, dirty_price, and yield_to_maturity, tested against Excel"
```

THE COPY

The copy replaces your files, so anything of your own must be in the work folder first.

`Copy-Item` copies files, here into **`.`**, the current folder; pytest shows 23 passed.

PRACTISE THE UNDO

To practise the undo, make the colleague's edit by hand, then run **`git commit -am "Compound the final period"`**, **`python -m pytest -q --tb=no`** (3 failed) and **`git revert --no-edit HEAD`**.

You Can Now

- Tell the quote from the invoice, and price the invoice in dollars.
- Discount from settlement over w + k periods, in Python and in a sheet.
- Price any settlement date with **`price`** and **`dirty_price`**, tied to Excel's **`PRICE`**.
- Show that a new function keeps the old answers on a coupon date.
- Price the final coupon period with simple interest, as Excel does.
- Solve a yield with **`yield_to_maturity`**, tied to **`YIELD`**, and say when **`YIELD`** does not undo **`PRICE`**.
- Undo a committed change with **`git revert`**.
- Put the accrued back on every trade of a blotter.

NEXT
WEEK 2 · DAY 4

Duration and DV01

How far a price moves when the yield moves: Macaulay and modified duration and DV01, per 100 and per position, tied to Excel's DURATION and MDURATION.